Economics S5040: Dynamic Stochastic Modeling

Supplement to Lecture 11: Extensions of the Solow Growth Model

Eric Fisher

Fall 2026

1 Introduction

Our original Solow model assumed a constant labor force and no technical progress. With Cobb–Douglas production, a constant saving rate s, and depreciation rate δ, capital per worker obeyed

$$k_{t+1} = sAk_t^{\alpha} + (1-\delta)k_t, \qquad 0 < \alpha < 1.$$

We now introduce three extensions:

1. Labor-force growth at rate n.
2. Labor-augmenting technical progress at rate g.
3. Linearization around the steady state and the half-life of convergence.

We retain the central Solow assumption: the saving rate is exogenous.

Throughout, n and g denote discrete-time net growth rates. Thus $1+n$ and $1+g$ are gross growth factors.

2 Extension 1: Labor-Force Growth

Suppose the number of physical workers grows according to

$$L_{t+1} = (1+n)L_t. \tag{1}$$

For now, retain the production function

$$Y_t = AK_t^{\alpha}L_t^{1-\alpha}.$$

Define capital and output per physical worker:

$$k_t = \frac{K_t}{L_t}, \qquad y_t = \frac{Y_t}{L_t} = Ak_t^{\alpha}.$$

Aggregate capital accumulation is unchanged:

$$K_{t+1} = sY_t + (1-\delta)K_t. \tag{2}$$

But capital per worker now satisfies

$$k_{t+1} = \frac{K_{t+1}}{L_{t+1}} \tag{3}$$

$$= \frac{sY_t + (1-\delta)K_t}{(1+n)L_t}. \tag{4}$$

Hence

$$\boxed{k_{t+1} = \frac{sAk_t^{\alpha} + (1-\delta)k_t}{1+n}.} \tag{5}$$

A steady state requires $k_{t+1} = k_t = k^*$, so

$$(1+n)k^* = sA(k^*)^{\alpha} + (1-\delta)k^*.$$

Rearranging,

$$\boxed{sA(k^*)^{\alpha} = (n+\delta)k^*.} \tag{6}$$

The positive steady state is therefore

$$\boxed{k^* = \left(\frac{sA}{n+\delta}\right)^{1/(1-\alpha)}.} \tag{7}$$

The economic interpretation is straightforward. Investment must accomplish two tasks:

1. Replace depreciated capital.
2. Equip additional workers with capital.

Thus labor-force growth reduces steady-state capital per worker, holding other parameters fixed.

3 Extension 2: Labor-Augmenting Technical Progress

Let E_t measure the efficiency of each physical worker. Assume

$$E_{t+1} = (1+g)E_t. \tag{8}$$

Total effective labor is $E_t L_t$.
We now write production as

$$\boxed{Y_t = K_t^{\alpha}(E_t L_t)^{1-\alpha}.} \tag{9}$$

For simplicity, the constant productivity multiplier is normalized to one.

3.1 Variables per Effective Worker

Define

$$\tilde{k}_t = \frac{K_t}{E_t L_t}, \qquad \tilde{y}_t = \frac{Y_t}{E_t L_t}. \tag{10}$$

Constant returns to scale implies

$$\boxed{\tilde{y}_t = \tilde{k}_t^{\alpha}.} \tag{11}$$

Because

$$E_{t+1}L_{t+1} = (1+g)(1+n)E_tL_t,$$

we can rewrite capital accumulation as

$$\tilde{k}_{t+1} = \frac{K_{t+1}}{E_{t+1}L_{t+1}} \tag{12}$$

$$= \frac{sY_t + (1-\delta)K_t}{(1+g)(1+n)E_tL_t}. \tag{13}$$

Consequently,

$$\boxed{\tilde{k}_{t+1} = \frac{s\tilde{k}_t^{\alpha} + (1-\delta)\tilde{k}_t}{(1+n)(1+g)}.} \tag{14}$$

This is the fundamental equation of the augmented Solow model.

The state variable is now capital per effective worker, rather than capital per physical worker.

3.2 The Steady State

At a steady state in effective units,

$$\tilde{k}_{t+1} = \tilde{k}_t = \tilde{k}^*.$$

Using equation (14),

$$(1+n)(1+g)\tilde{k}^* = s(\tilde{k}^*)^{\alpha} + (1-\delta)\tilde{k}^*.$$

Rearranging,

$$s(\tilde{k}^*)^{\alpha} = \left[(1+n)(1+g) - (1-\delta)\right]\tilde{k}^* \tag{15}$$

$$= (\delta + n + g + ng)\tilde{k}^*. \tag{16}$$

Therefore,

$$\boxed{\tilde{k}^* = \left(\frac{s}{\delta + n + g + ng}\right)^{1/(1-\alpha)}.} \tag{17}$$

Steady-state output per effective worker is

$$\boxed{\tilde{y}^* = (\tilde{k}^*)^{\alpha}.} \tag{18}$$

The expression

$$\delta + n + g + ng$$

is the exact discrete-time amount of investment per unit of effective capital required to keep capital per effective worker constant.

For small n and g, it is approximately

$$\delta + n + g.$$

3.3 The Balanced-Growth Path

At the steady state, capital and output per effective worker are constant.

But output per physical worker is

$$\frac{Y_t}{L_t} = E_t \tilde{y}_t. \tag{19}$$

On the balanced-growth path, $\tilde{y}_t = \tilde{y}^*$, so

$$\frac{Y_{t+1}/L_{t+1}}{Y_t/L_t} = \frac{E_{t+1}}{E_t} \tag{20}$$

$$= 1 + g. \tag{21}$$

Hence

$$\boxed{\text{Growth rate of real GDP per physical worker} = g.} \tag{22}$$

Aggregate output is

$$Y_t = E_t L_t \tilde{y}_t.$$

Along the balanced-growth path,

$$\frac{Y_{t+1}}{Y_t} = \frac{E_{t+1}}{E_t}\frac{L_{t+1}}{L_t} \tag{23}$$

$$= (1+g)(1+n). \tag{24}$$

Thus

$$\boxed{\text{Growth rate of aggregate real GDP} = n + g + ng.} \tag{25}$$

This is an exact discrete-time result.

The familiar expression $n + g$ is an approximation.

Capital per physical worker also grows at rate g, and aggregate capital grows at rate $n+g+ng$, along the balanced-growth path.

4 Extension 3: Linearization and Convergence

We now ask how rapidly the economy approaches its steady state.

Define the transition function

$$H(k) = \frac{sk^\alpha + (1-\delta)k}{(1+n)(1+g)}. \tag{26}$$

Then

$$\tilde{k}_{t+1} = H(\tilde{k}_t). \tag{27}$$

The steady state satisfies

$$H(\tilde{k}^*) = \tilde{k}^*.$$

4.1 A First-Order Taylor Approximation

Expand H around the steady state:

$$H(\tilde{k}_t) \approx H(\tilde{k}^*) + H'(\tilde{k}^*)(\tilde{k}_t - \tilde{k}^*). \tag{28}$$

Since $H(\tilde{k}^*) = \tilde{k}^*$,

$$\tilde{k}_{t+1} - \tilde{k}^* \approx H'(\tilde{k}^*)(\tilde{k}_t - \tilde{k}^*). \tag{29}$$

Define the deviation from steady state as

$$x_t = \tilde{k}_t - \tilde{k}^*.$$

Then the linearized system is

$$\boxed{x_{t+1} \approx \lambda x_t, \qquad \lambda = H'(\tilde{k}^*).} \tag{30}$$

4.2 Computing the Derivative

Differentiating,

$$H'(k) = \frac{s\alpha k^{\alpha-1} + 1 - \delta}{(1+n)(1+g)}. \tag{31}$$

At the steady state,

$$\lambda = \frac{s\alpha(\tilde{k}^*)^{\alpha-1} + 1 - \delta}{(1+n)(1+g)}. \tag{32}$$

From the steady-state condition,

$$s(\tilde{k}^*)^{\alpha-1} = \delta + n + g + ng.$$

Substitution gives

$$\boxed{\lambda = \frac{\alpha(\delta + n + g + ng) + 1 - \delta}{(1+n)(1+g)}.} \tag{33}$$

Equivalently,

$$\boxed{\lambda = 1 - \frac{(1-\alpha)(\delta + n + g + ng)}{(1+n)(1+g)}.} \tag{34}$$

For the usual parameter values,

$$0 < \lambda < 1.$$

The steady state is therefore locally stable.

Notice that the saving rate s does not appear in equation (34). With Cobb–Douglas production, saving changes the steady-state level of capital but not the local convergence factor.

4.3 The Speed of Convergence

Iterating the linearized equation gives

$$x_{t+1} \approx \lambda x_t,$$
$$x_{t+2} \approx \lambda^2 x_t,$$

and, in general,

$$\boxed{x_{t+h} \approx \lambda^h x_t.} \tag{35}$$

Thus deviations from the steady state decay approximately geometrically.

The closer λ is to one, the slower the convergence.

4.4 The Half-Life

Define the half-life $h_{1/2}$ as the number of periods required for the magnitude of a small deviation from the steady state to fall by one-half.

We require

$$\lambda^{h_{1/2}} = \frac{1}{2}.$$

Taking logarithms,

$$h_{1/2} \ln \lambda = \ln(1/2).$$

Hence

$$\boxed{h_{1/2} = \frac{\ln(1/2)}{\ln \lambda}.} \tag{36}$$

This is the half-life measured in model periods.

For an annual model, it is measured in years.

The formula applies when $0 < \lambda < 1$ and is a local approximation to the nonlinear system.

4.5 Numerical Example

Suppose the model has annual periods, with

$$\alpha = \frac{1}{3}, \qquad \delta = 0.05, \qquad n = 0.01, \qquad g = 0.02.$$

The gross growth factor of effective labor is

$$(1+n)(1+g) = 1.01(1.02) = 1.0302.$$

The effective-capital replacement term is

$$\delta + n + g + ng = 0.05 + 0.01 + 0.02 + 0.0002 = 0.0802.$$

Therefore,

$$\lambda = 1 - \frac{(1 - 1/3)(0.0802)}{1.0302} \tag{37}$$

$$\approx 0.9481. \tag{38}$$

The half-life is

$$h_{1/2} = \frac{\ln(0.5)}{\ln(0.9481)} \tag{39}$$

$$\approx 13.0 \text{ years}. \tag{40}$$

Thus, near the steady state, it takes approximately 13 years for a deviation of capital per effective worker to shrink by half.

After another 13 years, the remaining deviation is approximately halved again.

The calculation describes local convergence. Far from the steady state, the nonlinear system need not have the same exact half-life.

5 Summary: What to Remember

1. With labor-force growth,
$$L_{t+1} = (1+n)L_t.$$
Investment must replace depreciated capital and equip new workers with capital.

2. With labor-augmenting technical progress,
$$E_{t+1} = (1+g)E_t.$$
The appropriate state variable is capital per effective worker:
$$\tilde{k}_t = \frac{K_t}{E_t L_t}.$$

3. Capital per effective worker evolves according to
$$\boxed{\tilde{k}_{t+1} = \frac{s\tilde{k}_t^{\alpha} + (1-\delta)\tilde{k}_t}{(1+n)(1+g)}.}$$

4. At the steady state,
$$\boxed{s(\tilde{k}^*)^{\alpha} = (\delta + n + g + ng)\tilde{k}^*.}$$

5. On the balanced-growth path, real GDP per physical worker grows at the rate
$$\boxed{g}.$$

6. On the balanced-growth path, aggregate real GDP grows at the exact discrete-time rate
$$\boxed{n + g + ng}.$$
For small n and g, this is approximately $n + g$.

7. Linearization around the steady state gives
$$x_{t+1} \approx \lambda x_t,$$
where
$$\boxed{\lambda = 1 - \frac{(1-\alpha)(\delta + n + g + ng)}{(1+n)(1+g)}.}$$

8. The approximate half-life of convergence is
$$\boxed{h_{1/2} = \frac{\ln(1/2)}{\ln \lambda}.}$$

The central distinction: Capital and output per effective worker converge to constants. Capital and output per physical worker grow at rate g along the balanced-growth path, while aggregate capital and real GDP grow at rate $n + g + ng$.

During the transition to balanced growth, actual growth rates also reflect changes in capital and output per effective worker.